STEAM GAMING MARKET ANALYSIS AND REVIEW PREDICTION USING R

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***Abstract*—The digital gaming industry has grown significantly, creating opportunities to analyze game characteristics, pricing patterns, and player feedback using data science techniques. Understanding these patterns can help identify relationships between game attributes and user reception. This project presents a data science and machine learning approach for analyzing the Steam gaming market using the R programming environment. The study uses the Steam Games Dataset 2025, obtained from Kaggle. The dataset contains information about games, including available details related to pricing, release dates, genres, review statistics, and other game characteristics. Data preprocessing is performed to identify and handle duplicate records, missing values, invalid prices, and inconsistent data types. Exploratory data analysis is conducted to examine game release trends, pricing distributions, review categories, and genre popularity. Statistical analysis is used to investigate the association between game prices and positive review percentages and to compare review percentages between free and paid games. For predictive analysis, games are categorized into Positive and Non-Positive review classes using a predefined threshold of 70% positive reviews. Logistic Regression, Decision Tree, and Random Forest classifiers are developed and evaluated using a training and testing procedure. Model performance is assessed using accuracy, precision, recall, F1-score, and specificity. The models are compared using the same test dataset, and Random Forest feature importance is examined to understand the contribution of the selected predictors. The project demonstrates how R can be used to conduct exploratory analysis, statistical testing, and predictive modelling on real-world gaming data.**

***Index Terms*—Steam games, gaming market analysis, R programming, exploratory data analysis, statistical analysis, machine learning, review prediction, Random Forest.**

I. INTRODUCTION

The video game industry has become an important part of the digital entertainment market. Digital distribution platforms allow users to purchase, download, and review games across a wide variety of categories. Steam is one such platform, providing information about games, including their names, release dates, prices, genres, and user reviews.

The availability of gaming data creates opportunities for data-driven analysis. Developers and publishers may be interested in understanding which genres are frequently represented, how game prices vary, and how player reviews differ across products. Data analysts can use these records to identify trends and investigate relationships between game characteristics and user reception.

However, gaming datasets may contain missing values, duplicate entries, inconsistent data types, and unusual observations. These issues must be addressed before meaningful analysis can be performed. Exploratory data analysis and statistical testing can help identify patterns and relationships before machine learning models are developed.

Machine learning provides methods for identifying patterns in structured data and predicting outcomes from existing observations. Classification algorithms can categorize games according to their review percentages. Different algorithms may produce different results because they use different approaches to learning relationships between input variables and target categories.

This project applies R programming to analyze the Steam Games Dataset 2025. It combines data preprocessing, exploratory data analysis, statistical testing, and machine learning to examine gaming market characteristics and predict review categories. Three classification algorithms are compared to understand their relative performance.

The project provides a reproducible analytical workflow for transforming a raw dataset into visualizations, statistical findings, and predictive results. The findings are limited to the dataset and experimental configuration used and should not be interpreted as evidence that particular game characteristics cause positive reviews.

II. PROBLEM FORMULATION AND OBJECTIVES

A. Problem Formulation

The main problem addressed in this project is the analysis of Steam games and the classification of games into Positive and Non-Positive review categories using available numerical characteristics.

Let the dataset be represented as:

$$\mathcal{D} = \{(x_i, y_i)\}, fori = 1, 2, ..., N \tag{1}$$

where x represents the selected numerical features of game i, y represents its review category, and N represents the number of usable observations.

The target variable is constructed from the positive review percentage. A game with a positive review percentage of at least 70% is assigned to the Positive category, while a game with a percentage below 70% is assigned to the Non-Positive category.

This threshold is a project-defined classification rule rather than an official Steam classification.

The objective is to learn a relationship between the available numerical characteristics and the review category. The trained models are evaluated using a separate test dataset to estimate their classification performance on observations that were not used for model training.

B. Objectives

The main objectives of the project are:

To analyze a real-world Steam gaming dataset.

To identify and handle duplicate records, missing values, invalid prices, and data inconsistencies.

To perform exploratory data analysis using statistical summaries and visualizations.

To investigate the relationship between game prices and positive review percentages.

To compare review percentages between free and paid games using statistical testing.

To develop Logistic Regression, Decision Tree, and Random Forest classification models.

To evaluate the models using accuracy, precision, recall, F1-score, and specificity.

To identify a suitable classification model based on its observed performance.

To document the analytical workflow and results for academic reporting and reproducibility.

C. Scope of the Study

The study focuses on structured game information available in the selected Steam dataset. It includes data cleaning, exploratory data analysis, statistical testing, and binary classification using R.

The project is limited to the fields available in the selected CSV file and the predictors used in the implementation. It does not directly analyze review text using natural language processing, estimate actual game sales, or establish causal relationships between game characteristics and commercial success.

III. RELATED WORK

Data analysis and machine learning are widely used to understand consumer preferences and patterns in digital markets. In the gaming industry, structured datasets can be examined to investigate product characteristics, pricing, popularity, and user feedback.

Exploratory data analysis is an important initial stage of a data science project. Statistical summaries and graphical representations help identify distributions, unusual observations, and relationships among variables. For gaming data, these techniques can reveal changes in game releases, differences in pricing, and patterns in review percentages.

Logistic Regression is a statistical classification method that estimates the probability of an outcome based on input variables. It provides a useful baseline because its results can be interpreted through the relationship between predictors and the target category.

Decision Tree classification uses feature-based rules to divide observations into groups. Its structure can be relatively easy to interpret, although its performance may depend on the training data and the selected tree complexity.

Random Forest is an ensemble learning algorithm that combines predictions from multiple decision trees. The use of multiple trees can capture nonlinear relationships and may improve predictive performance compared with an individual tree. Random Forest also provides feature-importance estimates that can assist in interpreting the fitted model.

The present project combines these methods to examine Steam gaming data. Rather than assuming that a particular algorithm will perform best, all three classifiers are evaluated using the same held-out test dataset and compared using multiple performance metrics.

IV. DATASET DESCRIPTION

The dataset used in this project is the Steam Games Dataset 2025, obtained from Kaggle.

Dataset source: https://www.kaggle.com/dsv/11017460

The dataset contains records describing games available on Steam. Depending on the downloaded CSV version, the available fields may include game identifiers, names, release dates, prices, genres, review statistics, required age, downloadable content count, Metacritic scores, peak concurrent users, and achievement counts.

Only variables available in the selected CSV file are used in the analysis.

A. Dataset Summary

TABLE I
DATASET SUMMARY

Parameter	**Value**
Dataset name	Steam Games Dataset 2025
Source	Kaggle
File format	CSV
Initial records	[89,618]
Initial variables	[47]
Records after cleaning	[53,199]
Target variable	Review category
Classification classes	Positive and Non-Positive
Programming environment	RStudio

B. Target Variable Construction

The target variable is created from the positive review percentage. Games with a positive review percentage of 70% or higher are labelled Positive, while games with a percentage below 70% are labelled Non-Positive.

The transformation converts the original percentage into a binary classification outcome. The distribution of the two categories is examined to understand whether the target classes are balanced.

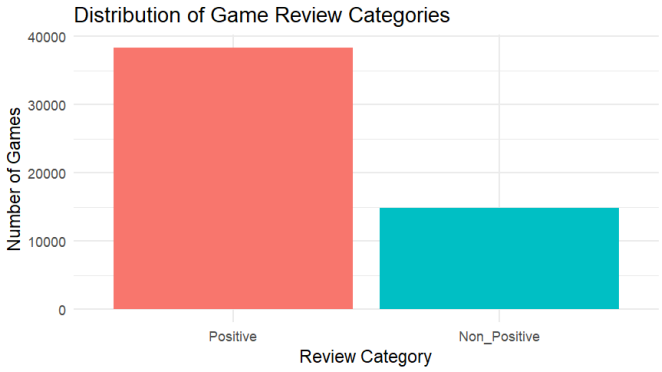


Fig. 1. Distribution of games in each review category.

V. PROPOSED METHODOLOGY

The proposed methodology consists of data loading, preprocessing, exploratory data analysis, statistical testing, feature selection, model development, evaluation, and interpretation. The workflow is implemented using R and associated data analysis and machine learning packages.

A. Data Loading

The CSV file is imported into RStudio using the data.table package. The initial dataset dimensions, column names, first few records, and summary statistics are inspected to understand the structure and contents of the dataset.

The janitor package is used to standardize column names, making them easier to reference during the subsequent stages of analysis.

B. Data Cleaning and Preprocessing

Data cleaning is performed before exploratory and predictive analysis. The main operations include:

1. Identifying and removing exact duplicate rows.
2. Removing duplicate game identifiers when an application ID is available.
3. Converting numerical variables into appropriate numeric data types.
4. Identifying invalid or negative game prices.
5. Extracting release years from available release dates.
6. Handling missing values in selected numerical predictors using median imputation.
7. Removing columns with excessive missing values according to the implemented threshold.
8. Removing records without a valid target category.
9. Excluding variables that directly determine the target from the predictor set to reduce target leakage.

The review percentage is obtained from an existing percentage field or calculated using positive and negative review counts when the required fields are available.

The preprocessing decisions are documented so that their effects on the final modelling dataset can be understood.

C. Exploratory Data Analysis

Exploratory data analysis is performed to understand the characteristics and distributions of the gaming data.

1) Game Release Trends: A line chart presents the number of recorded game releases in each year. It provides an overview of how the number of games in the dataset varies over time.

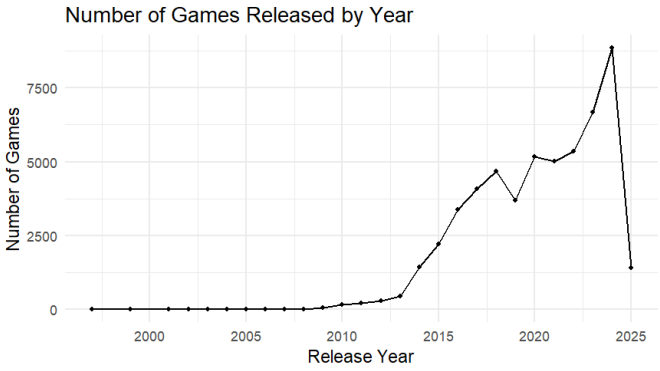


Fig. 2. Distribution of recorded game releases across years.

2) Game Price Distribution: A histogram is used to examine the distribution of game prices. The displayed range is limited to the 95th percentile to make the main price distribution easier to observe.

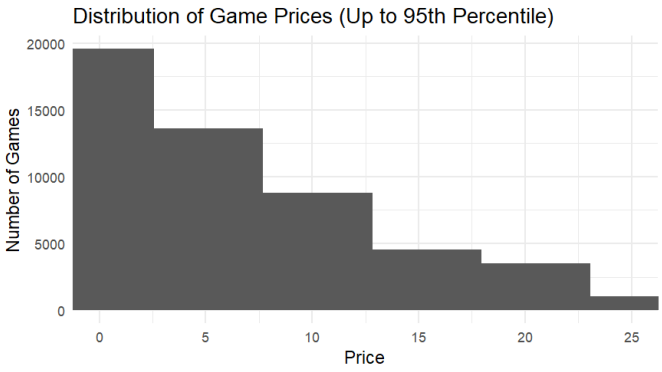


Fig. 3. Distribution of game prices within the displayed range.

3) Price and Review Relationship: A scatter plot displays game prices against positive review percentages. A fitted linear trend line provides a visual summary of the relationship between these variables.

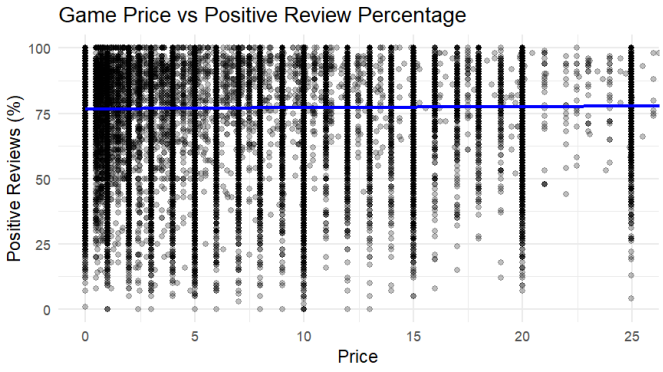


Fig. 4. Relationship between game prices and positive review percentages.

4) Genre Distribution: Where genre information is available, individual genre labels are separated and the ten most frequently represented genres are displayed using a horizontal bar chart.

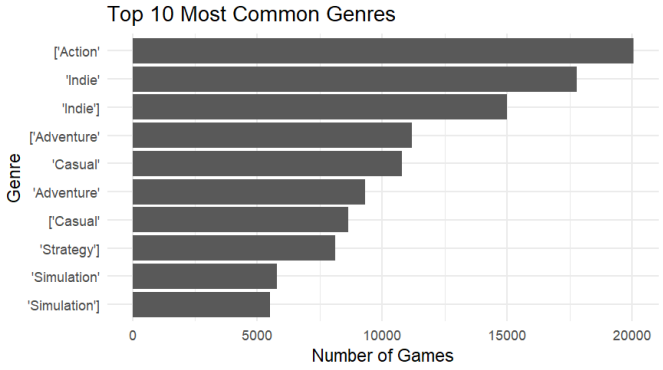


Fig. 5. Ten most frequently represented genres in the dataset.

D. Statistical Analysis

Statistical tests are used to investigate selected relationships in the dataset.

1) Spearman Correlation Test: Spearman's rank correlation test examines the association between game price and positive review percentage. It evaluates whether the variables have a monotonic relationship without requiring that the relationship be linear.

The null hypothesis is:

Null hypothesis: There is no monotonic association between game price and positive review percentage.

The alternative hypothesis is:

Alternative hypothesis: There is a monotonic association between game price and positive review percentage.

The R output gave Spearman's rho = 0.0500 with $p < 2.2 \times 10^{1}$. This indicates a very weak positive monotonic association between price and positive review percentage. Although statistically significant, the coefficient is close to zero, so the association is weak in practical terms.

2) Wilcoxon Rank-Sum Test: The Wilcoxon rank-sum test compares positive review percentages between free and paid games.

The null hypothesis is:

Null hypothesis: The review percentage distributions of free and paid games do not differ systematically.

The alternative hypothesis is:

Alternative hypothesis: The review percentage distributions of free and paid games differ.

A significance level of 0.05 is used for interpretation. The Wilcoxon rank-sum test returned W = 226,434,165 and $p < 2.2 \times 10^{1}$. Therefore, the test provides evidence that free and paid games have different review-percentage distributions. The console output does not report the direction or size of this difference, so those should not be inferred from the p-value alone.

E. Feature Selection and Model Preparation

The selected numerical features may include price, release year, required age, downloadable content count, Metacritic score, peak concurrent users, and achievement count, depending on their availability.

The review percentage and positive and negative review counts are excluded from the predictor set when they are used to construct the target variable. This prevents the model from using information that directly reveals the classification outcome.

The modelling dataset is limited to a maximum of 20,000 observations to keep the computational requirements manageable. The data is divided into training and testing subsets using an 80:20 split with stratified sampling.

Numerical features are standardized using parameters calculated from the training data. The same transformation is then applied to the test data.

F. Machine Learning Models

Three classification algorithms are developed and evaluated.

1) Logistic Regression: Logistic Regression estimates the probability that a game belongs to a particular review category. The predicted probability is converted into a class label using a threshold of 0.5. The model provides a baseline for comparing the performance of the other classifiers.

2) Decision Tree: The Decision Tree classifier divides observations using feature-based decision rules. The implementation restricts the tree depth and applies a complexity parameter to control tree growth.

3) Random Forest: Random Forest combines the predictions of multiple decision trees. The implementation uses 100 trees and generates a feature-importance plot to examine the relative contribution of the selected predictors.

All three models are evaluated using the same held-out test dataset.

G. Model Evaluation

The models are assessed using the following metrics:

Accuracy: Proportion of test observations classified correctly.

Precision: Proportion of predicted Positive games that are actually Positive.

Recall: Proportion of actual Positive games correctly identified.

F1-score: Harmonic mean of precision and recall.

Specificity: Proportion of actual Non-Positive games correctly identified.

These metrics provide a more complete assessment than accuracy alone, particularly if the two review categories are imbalanced.

VI. RESULTS AND DISCUSSION

This section reports the numerical findings produced by the R implementation. The original CSV contained 89,618 rows and 47 variables. No exact duplicate rows were found, and the dimensions remained 89,618 × 47 after duplicate removal. Valid review categories were available for 53,199 games; 36,419 records had missing review percentages and were excluded from the target-based analysis.

A. Dataset Preparation Results

After target filtering and preprocessing, the exploratory analysis dataset contained 53,199 games. The modelling dataset was then sampled to 20,000 observations and contained seven numerical predictors plus the target variable (20,000 rows × 8 columns).

The preprocessing stage addressed duplicate records, missing values, invalid prices, and data-type inconsistencies according to the implemented rules. Records without a valid target category were excluded from the modelling dataset.

The final modelling sample contained 20,000 observations: 14,458 Positive games (72.29%) and 5,542 Non-Positive games (27.71%). The stratified split produced 16,001 training records and 3,999 test records.

B. Exploratory Data Analysis Results

Among the 53,199 games with valid review categories, 38,314 were Positive (72.02%) and 14,885 were Non-Positive (27.98%). The classes were therefore imbalanced, with Positive games forming the majority.

The valid release years ranged from 1997 to 2025, with a median release year of 2021. Game prices ranged from 0 to 199.99; the median price was 4.99 and the mean was 7.869. The review percentage had a median of 81% and a mean of 77.08%. The price-versus-review plot should be interpreted alongside the Spearman result, which showed only a very weak positive association (rho = 0.0500).

The R script generated a top-10 genre plot. The console log supplied here does not print the individual genre counts, so the exact leading genres should be reported by reading Figure 5 rather than guessing them.

These findings provide a descriptive overview of the available games and their characteristics. They should be interpreted in the context of the dataset snapshot rather than as a complete representation of the entire gaming industry.

TABLE II
COMPARISON OF CLASSIFICATION MODELS

Model	Accuracy Precision Recall F1-score Specificity
Logistic Regression	[0.7209302] [0.7225984] [0.9965410] [0.8377435] [0.001805054
Decision Tree	[0.7229307] [0.7229307] [1.0000000] [0.8391872] [0.000000000]
Random Forest	[0.7319330] [0.7462767] [0.9533034] [0.8371810] [0.154332130]

C. Statistical Test Results

The Spearman correlation test produced rho = 0.05002996 and $p < 2.2 \times 10^{1}$. The association between price and positive review percentage is positive but very weak. The small p-value indicates statistical evidence of a non-zero association in this dataset; it does not mean the relationship is strong or causal.

The Wilcoxon rank-sum test returned W = 226,434,165 and $p < 2.2 \times 10^{1}$. At the 0.05 significance level, the results support a difference between the review-percentage distributions of free and paid games. The output shown does not establish which group has the higher distribution or the size of the difference.

The findings describe associations within the analyzed dataset and do not establish causal relationships.

D. Model Comparison

The three classification models were evaluated using the same test dataset.

The Decision Tree achieved the highest F1-score (0.8392), with accuracy 0.7229, precision 0.7229, recall 1.0000, and specificity 0.0000. However, it predicted every test record as Positive, so it failed to identify any Non-Positive games. Random Forest achieved the highest accuracy (0.7319) and the highest specificity (0.1543) of the three models, but its specificity was still low. Thus, the best F1-score alone does not identify a practically balanced classifier.

The final model should be selected by considering all relevant metrics and the class distribution, rather than relying exclusively on accuracy.

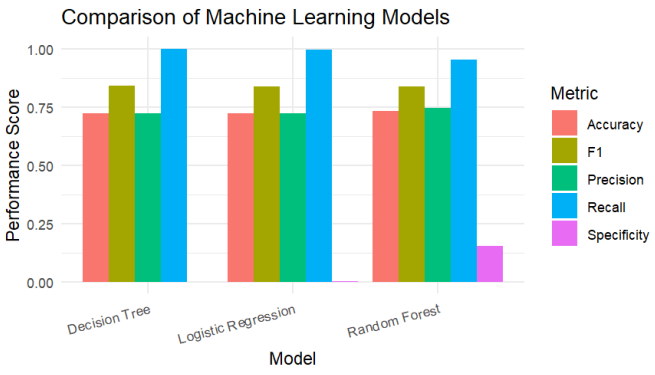


Fig. 6. Comparison of classification model performance.

E. Feature Importance

The Random Forest model generates a feature-importance plot showing the relative importance of the selected numerical predictors.

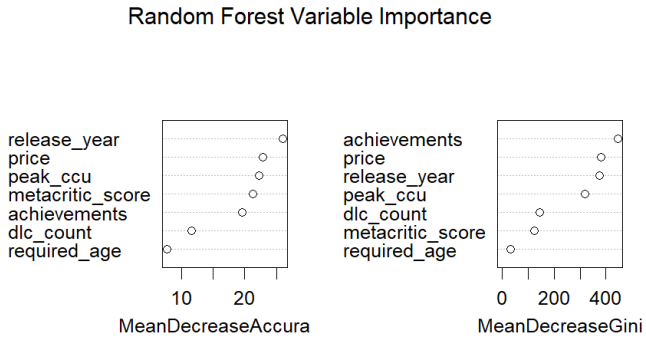


Fig. 7. Comparison of classification model performance.

The R script generated a Random Forest variable-importance plot, but the console text does not include the feature-importance ranking values. Insert Figure 7 and name the leading predictors only after checking that plot.

Feature importance does not demonstrate that a particular variable causes positive reviews. The ranking may also be influenced by relationships among predictors and characteristics of the dataset.

F. Overall Discussion

The project demonstrates how R programming can be used to analyze gaming data and develop classification models. The combination of visualization and statistical analysis provides an overview of the dataset before predictive modelling.

Logistic Regression provides a baseline, Decision Tree offers interpretable decision rules, and Random Forest can represent more complex relationships through an ensemble of trees.

The results reveal an important limitation: Logistic Regression and Decision Tree have near-zero specificity, and the Decision Tree predicts every test observation as Positive. Random Forest performs somewhat better at identifying Non-Positive games, but its specificity is only 0.1543. The class imbalance and weak identification of the minority class mean that none of the models should be described as reliably balanced. Further work should consider class weights or resampling applied only to training data, threshold tuning, and additional evaluation such as balanced accuracy and class-wise metrics. The findings should not be generalized to all Steam games or future releases without further validation.

VII. Statistical Analysis and Interpretation

The statistical analysis complements the predictive modelling by examining relationships between selected variables.

The Spearman correlation test investigates whether game price and positive review percentage are associated in a monotonic pattern. A positive coefficient indicates that higher prices tend to be associated with higher review percentages in the observed data, while a negative coefficient indicates the opposite pattern.

The Wilcoxon rank-sum test examines whether the review percentage distributions of free and paid games differ. Its result should be interpreted together with the observed distributions and the magnitude of the difference.

The classification metrics provide a separate evaluation of predictive performance. Accuracy measures overall correctness, while precision, recall, F1-score, and specificity describe different aspects of classification performance.

These analyses answer different questions: statistical tests investigate associations in the observed data, whereas classification metrics evaluate predictions on the held-out test set. Differences in model scores should not automatically be interpreted as statistically significant differences in generalization performance.

VIII. Limitations

The project has several limitations.

First, the analysis depends on the fields available in the selected dataset. Missing or incomplete information may limit the number of usable predictors.

Second, the dataset represents a particular snapshot of Steam games. The observed patterns may not reflect future releases or changes in player preferences.

Third, the target variable is constructed using a predefined threshold of 70% positive reviews. Different thresholds may produce different class distributions and model performance.

Fourth, the implementation uses numerical predictors and does not analyze detailed game descriptions or player-review text.

Fifth, the modelling sample is limited to a maximum of 20,000 records when more observations are available. Consequently, results may differ from those obtained using the entire dataset.

Sixth, the evaluation uses a single training and test split. Performance may vary across different random splits, and further validation is needed to establish the stability of the models.

Finally, the project predicts review categories rather than actual game sales, profitability, or long-term commercial success. Positive reviews alone are not sufficient to establish the overall success of a game.

IX. Future Work

Several extensions could improve the project.

First, additional game characteristics could be included if available and relevant to the classification task.

Second, alternative classification thresholds could be evaluated to determine how the definition of positive reviews affects class balance and model performance.

Third, hyperparameter tuning and repeated cross-validation could be used to obtain more reliable estimates of predictive performance.

Fourth, additional algorithms, such as Support Vector Machines and Gradient Boosting, could be evaluated against the current models.

Fifth, text mining and sentiment analysis could be applied to player reviews if review text is available. This would provide information beyond numerical game characteristics.

Sixth, explainability techniques could be used to better understand individual predictions and the contribution of different features.

Finally, the project could be extended to analyze changes over time or predict other outcomes, such as review volume. Each extension would require a clearly defined target, suitable predictors, and an appropriate evaluation procedure.

X. CONCLUSION

This project presented a data science and machine learning workflow for Steam gaming market analysis using RStudio. The study included dataset inspection, data cleaning, exploratory data analysis, statistical testing, and the development of Logistic Regression, Decision Tree, and Random Forest classifiers. The review category was constructed using a threshold of 70% positive reviews. The resulting binary classification task provided a basis for comparing the predictive performance of the selected algorithms. The exploratory analysis examined release trends, game prices, review patterns, and genre distributions. Statistical tests were used to investigate the association between game price and positive review percentage and to compare review percentage distributions between free and paid games. The final model selection will be based on the actual test-set metrics, including accuracy, precision, recall, F1-score, and specificity. The Random Forest feature-importance plot provides additional information about the contribution of the selected numerical predictors. Overall, the project demonstrates how R can transform structured gaming data into interpretable visualizations, statistical findings, and predictive models. The results provide a foundation for further work involving improved feature engineering, more extensive validation, and broader gaming market analysis.

REFERENCES

[1] Kaggle, "Steam Games Dataset 2025," dataset repository. [Online]. Available: https://www.kaggle.com/dsv/11017460. Accessed: Oct. 26, 2026.

[2] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, *Applied Logistic Regression*, 3rd ed. Hoboken, NJ, USA: Wiley, 2013.

[3] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001.

[4] L. Breiman, J. H. Friedman, R. A. Olshen, and C. J. Stone, *Classification and Regression Trees*. Belmont, CA, USA: Wadsworth, 1984.

[5] R Core Team, "R: A Language and Environment for Statistical Computing." R Foundation for Statistical Computing. [Online]. Available: https://www.r-project.org/.

[6] M. Kuhn, "Building predictive models in R using the caret package," *Journal of Statistical Software*, vol. 28, no. 5, pp. 1–26, 2008.